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SHORT-PAPER

AI and Gamification for Employees with ADHD: A Tool for Enhancing Workplace Focus & Productivity

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Abstract

Attention-deficit hyperactivity disorder (ADHD) is a neurodevelopmental condition that significantly affects daily functioning, academic and workplace productivity. The unique challenges that individuals with ADHD face in professional settings are often irresolvable through traditional treatments, thereby limiting these populations' ability to focus, manage tasks and maintain efficiency at work. To address this problem, we developed an AI-driven gamification tool called WADAQ, which is designed to enhance cognitive and executive functioning. Using machine learning (ML) models, WADAQ provides a personalised and engaging approach to ADHD support. Initially, classification models assess and predict ADHD likelihood with high accuracy, after which users undergo subtype assessment to enable the system to recommend tailored mini-games on the basis of reinforcement learning (RL) techniques. These features are intended to help employees develop key workplace skills, such as sustained attention, impulse control and task management. An interactive dashboard enables real-time performance tracking for optimized training. The initial results indicated that when WADAQ leverages a support vector machine (SVM), it achieves 97% predictive accuracy. This accuracy is maintained after hyperparameter optimisation with GridSearchCV. By integrating AI and gamification, WADAQ serves as a scalable, adaptive solution aimed at improving workplace performance. WADAQ provides a scalable, AI-driven ADHD intervention that informs digital therapeutics and workplace policies, enabling inclusive hiring, tailored

accommodations, and improved employment outcomes for individuals with ADHD.

CCS Concepts

• **Applied computing** → **Life and medical sciences**; *Health care information systems*.

Keywords

ADHD, Game, Machine Learning, Reinforcement Learning, Accessibility, Employment

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1 Introduction

Traditional treatments for ADHD rely primarily on behavioural interventions and pharmacological solutions, but these approaches are often confronted with limitations, such as side effects and inconsistent effectiveness. This problem highlights the need for personalised, engaging alternatives that better address ADHD symptoms in real-world settings, including the workplace. ADHD is associated with impairments in attention, impulse control and executive functioning resulting from structural and functional brain differences and neurotransmitter imbalances, which complicate symptom management in daily life [1]. To address these challenges, this study developed WADAQ, an AI-driven gamification tool designed to enhance ADHD treatment while improving workplace performance. WADAQ integrates machine learning (ML) and reinforcement learning (RL) technologies to personalise gameplay,

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adapting to individual needs and optimising cognitive skill development. Beyond enabling symptom management, WADAQ focuses on skill transferability, ensuring that users can apply learned strategies to real-life tasks, particularly in professional contexts. Through an initial ADHD assessment and targeted mini games, the system provides real-time feedback and allows for progress tracking, reinforcing sustained improvement. This research explored how ML and RL can enhance the accuracy of ADHD treatment (theoretical contribution), how game design can support different ADHD subtypes (practical contribution) and how employment-focused accessibility can improve workplace inclusion and job performance among individuals with ADHD (policymaking contribution).

2 Related Work

Data science (DS) combines statistics and computer science, and it involves the analysis of data behaviour using descriptive, diagnostic, predictive and prescriptive methods, helping organisations gain actionable insights on issues such as employee productivity, hiring biases, workplace accommodations, and the effectiveness of digital therapeutics in supporting neurodivergent individuals [2].

Meanwhile, gamification, which integrates game design into nongame contexts, boosts engagement, motivation and performance [3]. In healthcare, gamification has proven effective in improving treatment compliance and accessibility. Serious games for ADHD, such as Plan-It Commander, have fostered improvements in key skills, such as task management and working memory—abilities that are essential for workplace success. Both data science and gamification were leveraged in this research to create an AI-powered tool that personalises ADHD treatment, accordingly allowing the enhancement of focus, productivity and workplace engagement. Using predictive analytics and RL, the tool adapts to individual needs, thus improving workplace performance and reducing ADHD symptoms [4–6].

2.1 ADHD and Data Science Applications

DS enhances ADHD treatment through ML and interactive gaming. A study [7] predicted ADHD in children by using various ML models, such as random forests (RFs), decision trees (DTs), support vector machines (SVM) and K-nearest neighbours (KNNs), among which RFs achieved 85.5% accuracy. Despite the value offered by this study, however, its relevance is affected by the outdated dataset used by the researchers. Another investigation [8] combined EEG data with ML (SVM) in the game Focus, which achieved 98% accuracy but struggled with generalisation due to noisy data. Improvements can include the enhancement of noise management via DTs and enhanced preprocessing. Researchers have also employed game data analysis with models such as AdaBoost, JRip and RF in the game Groundskeeper, realising 78% accuracy but facing challenges related to sample size and model generalisability [9].

The abovementioned studies demonstrated the potential of ML and gaming as means by which to enhance ADHD diagnosis in spite of the researchers encountering generalisation issues and data noise. These challenges are overcome by our proposed model, WADAQ, which uses recent data, improved model methods and more individualised strategies. It also encompasses RL, thus personalising gameplay and elevating predictive accuracy. As a tool,

WADAQ not only aids in ADHD treatment but also amplifies workplace performance and productivity by tailoring interventions to individual needs.

2.2 ADHD and Games

Studies have shown that gamification and ML can significantly enhance ADHD treatment by improving engagement and cognitive functioning. For instance, researchers found that serious games, including Plan-It Commander, help individuals reinforce competencies such as task management, responsibility and working memory, which are essential to ensuring workplace performance [10]. A study explored the use of game-based interventions to reduce ADHD symptoms, demonstrating positive effects on attention and impulse control [11]. Furthermore, game-based assessments, including those used in Focus and Groundskeeper, have shown potential for predicting ADHD symptoms through behavioural data analysis and ML modelling. The problem is that these games are frequently constrained by generalisation issues, especially in diverse populations. Consequently, managing ADHD in employees can improve workplace performance and support successful employment [12]. Gamification boosts informal workplace learning, with its impact shaped by context and contributing to overall performance [13].

These limitations are overcome by WADAQ because it integrates gamification and ML in providing a personalised treatment experience, offering real-time feedback and adaptive gameplay that is customised to each user. By focusing on ADHD subtypes and integrating RL into the system, WADAQ not only enhances ADHD treatment but also powerfully augments workplace concentration, productivity and cognitive abilities, thereby enabling employees afflicted with ADHD to perform at their best [12,13].

3 Methodology

The proposed tool is an improvement over previously developed games, as it offers better accessibility for all ages through tablet availability, ensuring long-term relevance in workplace settings. It captures real-time data, which serve as bases for adjusting game difficulty on the grounds of user performance, thus addressing a broader range of ADHD symptoms. Using DTs to enhance feature selection reduces noise and dimensionality, elevating treatment effectiveness.

3.1 Games design

This section explores games designed to address the impulsivity/hyperactivity and inattention symptoms of ADHD [14], analysing their suitability for use in WADAQ. Three games intended to address impulsivity/hyperactivity were reviewed: The stop task game is aimed at improving response inhibition, but its effectiveness has been minimally examined in credible research, making it unsuitable for inclusion in WADAQ-driven intervention [15,16]. The balloon analogue risk task helps players make risk-based decisions, minimising impulsive behaviour. It was considered suitable for incorporation into WADAQ due to its focus on decision-making. Finally, the Fin game enhances impulse control and decision-making in a fast-paced format and is easily adaptable to WADAQ with AI integration and level tracking [17].

For inattention, the Orbit Path game requires sustained attention but is deemed excessively difficult for users, with severe level, rendering it unsuitable for WADAQ inclusion [18]. The Falling Dots game, which improves focus and hand–eye coordination, was also regarded as an excellent fit for WADAQ, enabling short bursts of engagement and providing immediate feedback [18]. The Through game challenges players with respect to decision-making and task management, and the game adapts performance levels and improves with AI integration, making it appropriate as an intervention tool within WADAQ. Lastly, Sudoku improves concentration but was evaluated as overly complex for young users and therefore in correspondent with WADAQ [18]. In conclusion, the games selected for inclusion in WADAQ are Fin, balloon analogue risk task, Falling Dots and Through—which lend themselves to AI integration, feedback delivery and reward provision.

3.2 Vanderbilt ADHD Rating Scale

The Vanderbilt ADHD Diagnostic Parent Rating Scale (VADPRS) is a comprehensive tool for assessing ADHD. Based on DSM-IV criteria, it comprises 55 questions meant to evaluate core ADHD behaviours, such as inattention, impulsivity and hyperactivity. It categorises patients into three subtypes: inattentive, hyperactive/impulsive and combined. The VADPRS is valuable for improving diagnostic accuracy and aiding treatment planning, making it a reliable tool for ADHD evaluation and management [19,20]. In relation to WADAQ, this scale provides a solid foundation for assessing ADHD subtypes, enabling the proposed system to personalise treatment. Its integration into WADAQ guarantees customised interventions that address specific ADHD symptoms.

4 Enhancing the accuracy of ADHD treatment with serious games

The succeeding sections outline the steps in enhancing the accuracy of ADHD treatment with serious games, starting with the selection of data sources, followed by exploratory data analysis (EDA) and the preprocessing necessary to improve model reliability.

4.1 Data Source

WADAQ data were derived from the 2022 National Survey of Children's Health (NSCH), which was administered to 54,104 participants and revolves around 490 characteristics related to behaviours, family history and health records. Although the dataset covers children up to 17 years old, the variables chosen for incorporation into our solution are relevant and applicable to adults as well. The dataset was used to create a pretest intended to detect ADHD, with the target variable, K2Q31A, indicating whether a healthcare provider diagnosed an individual with ADHD [21]. This information is essential for prediction, with a value of 1 pointing to ADHD diagnosis and a value of 2 indicating an ADHD-free individual [21].

4.2 Exploratory Data Analysis (EDA)

EDA is the first step in an ML project, with focus directed towards understanding key data characteristics through visualisation and statistical methods [22]. Univariate visualisation is conducted to analyse individual variables, in which skewed distribution is identified for the target variable (K2Q31A; majority ADHD-free) and a

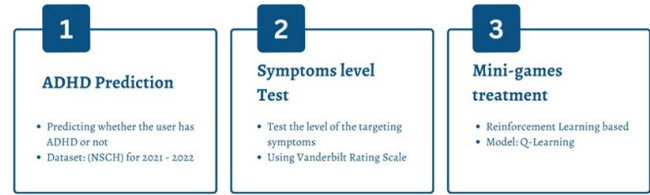


Figure 1: Wadaq's Proposed Model

right-skewed age distribution is detected with young individuals. Bivariate visualisation involves the exploration of relationships between two variables, with heatmaps showing strong correlations between ADHD presence and other features [23]. Multivariate visualisation entails examining multiple variables, revealing a weak positive correlation between SCREENTIME and MAKEFRIEND [24] (Appendix A).

4.3 Data Preprocessing

Preprocessing was essential to this research, as it ensured the integrity and usability of the NSCH dataset [21].

4.3.1 Duplicate Values. Removing duplicates ensures data quality and integrity. Our dataset was checked and found to have no duplicates, supporting accurate and reliable analyses [25].

4.3.2 Missing Values. Missing data can lead to biased estimates and errors [26]. Logistic regression (LR) was carried out for imputation, as it predicts missing values based on feature relationships and improves accuracy over simpler methods [27]. Columns with over 40% missing data were discarded, while LR imputed missing values in the remaining data, ensuring better quality for analysis.

4.3.3 Under-Sampling. Imbalanced datasets can cause the generation of biased models that favour the majority class (ADHD-free: 48,220 vs. ADHD presence: 5,519), reducing predictive accuracy for ADHD cases [28]. To address this issue, we applied random undersampling, selecting a subset of 5,519 ADHD-free instances to balance the dataset. This approach ensures equal representation, improving the model's effectiveness in generalising and making accurate predictions for both ADHD and ADHD-free groups. The upshot of all these is more reliable results.

4.4 Post-Processing Exploratory Data Analysis

Following data preparation, another EDA (univariate visualisation after preprocessing and bivariate visualisations) was conducted to examine the dataset after it was preprocessed and refined [29]. This step was crucial for evaluating the impact of the data preprocessing and ensuring data quality (Appendix A).

5 The Proposed Model (WADAQ)

5.1 Model Planning

This section outlines the planning and feature engineering for WADAQ (Appendix B), detailing the use of RL in the administration of the VADPRS to detect ADHD subtypes. It also provides an overview of the performance dashboard and user interface design of the proposed tool.

5.1.1 Feature Engineering. This process involves selecting relevant data and creating new variables to enhance model accuracy by reducing unnecessary features. In this project, from the 490 features in the dataset, 15 relevant variables were selected using a chi-square test to identify the most important categorical variables related to the target variable—ADHD diagnosis. This approach helped eliminate irrelevant or redundant features, prevent overfitting, reduce complexity and improve efficiency [30].

5.1.2 Selection of Classification Model. The classification models considered for WADAQ to predict ADHD occurrence or absence included DT [7], RF [7], KNN [7] and SVM [8,31]. These models were selected based on their robustness and past research on them [7,8,13], with DT offering interpretability, RF enhancing predictive accuracy through ensemble methods, KNN leveraging proximity for classification and SVM excelling in high-dimensional data classification. The models were chosen to handle the specific challenges confronting the use of the NSCH dataset and ADHD detection.

5.1.3 Reinforcement Learning. WADAQ implements RL using a model-free approach, specifically Q-learning, to personalise gameplay by continuously adjusting difficulty and mechanics based on real-time user performance. This dynamic adaptation is driven by a custom reward system, wherein an agent learns optimal actions through interaction, improving user experience while fostering skills in attention and behavioural regulation [32–34].

5.1.4 Symptom Scaling Tool. Based on the VADPRS, WADAQ focuses on questions related to the inattentive (1–9) and impulsive/hyperactive (10–18) subtypes to capture key ADHD symptoms. The scale uses a Likert system, with scores ranging from 0 (never) to 3 (very often). For diagnosis, DSM-IV criteria require a score of 2 or 3 on at least six questions within each subtype. These cutoff scores help determine whether the frequency and severity of symptoms warrant an ADHD diagnosis [35].

5.1.5 Accessibility. Accessibility ensures that information, activities and environments are inclusive and usable for all, particularly individuals with disabilities [36].

6 Accessibility Requirements

The design of the proposed tool accords with multiple WCAG 2.1 [37] standards to ensure accessibility, particularly for employees with ADHD, fostering a more inclusive and supportive workplace while also enhancing job interview performance. To reduce cognitive overload and improve focus, a limited colour palette is used in the user interfaces, consistent with WCAG 1.4.1 (Use of Colour). Text–background contrast was optimised for readability (WCAG 1.4.3: Contrast Minimum), while high-contrast icons were used to improve visibility and distinguishability (WCAG 1.4.11: Non-text Contrast). To accommodate varying attention spans and processing speeds, a prominently displayed adjustable timer allows users to control task duration, in accordance with WCAG 2.2.1 (Timing Adjustable).

For ease of use, intuitive navigation was implemented through the use of structured headings and labels (WCAG 2.4.6: Headings and Labels), while clear instructions accompany all interactive elements, with an additional 'help' button providing further guidance

(WCAG 3.3.2: Labels or Instructions). The game structure follows a logical sequence, preventing unnecessary cognitive strain (WCAG 1.3.2: Meaningful Sequence). Furthermore, a language toggle allows users to select either Arabic or English, ensuring accessibility for diverse workplaces and teams (WCAG 3.1.1: Language of Page).

For further support, WADAQ has a User Performance Dashboard that enables employees and employers to track real-time progress. The dashboard displays key metrics, such as average time, correctness rate (%), error rate (%) and best time. These data clear the way for providing immediate feedback, helping employees with ADHD improve their focus, task completion efficiency and cognitive performance over time [38,39].

7 Initial Results

Initial results for the classification models were obtained using two validation methods: hold-out validation and GridSearchCV. Using the hold-out method, the dataset was divided into training and testing subsets. The findings indicated that when WADAQ leverages achieved RF, DT, SVM and KNN in prediction, it achieved 94%, 94%, 85% and 91% accuracy, respectively. These results provided a baseline for assessing model performance. To enhance accuracy, GridSearchCV was deployed in optimising hyperparameters. This resulted in improved accuracy for most of the models, with RF achieving 94% accuracy (a 0.06% increase), DT increasing to 94% (a 0.06% increase), SVM with 97% (a 12% increase) and KNN slightly increasing to 92% (a 1% increase).

8 Conclusion

WADAQ is more than simply a game; it serves as a crucial tool that fosters workplace inclusivity for individuals with ADHD. Underlain by a structured yet engaging approach, WADAQ supports usage, from initial assessment to ongoing cognitive development. The platform begins with a user-friendly pretest to evaluate ADHD likelihood, followed by VADPRS administration for those who test positive, ensuring accurate subtype classification (inattention, impulsivity/hyperactivity or combined). Personalised mini-games are then recommended based on a user's ADHD profile, dynamically adjusting difficulty levels to enhance cognitive skills related to attention and impulse control. Users receive real-time feedback through a performance dashboard, allowing for improvement tracking over time.

Beyond enabling symptom management, WADAQ is an instrumental tool in professional contexts because it can be integrated into friendly interview processes, helping employers better understand a candidate's ADHD-related challenges and strengths. By offering insight into attention and impulse control patterns, WADAQ advances the efforts of talent managers to implement workplace accommodations that optimise productivity and well-being. Additionally, WADAQ provides ongoing support by refining cognitive functions essential for job success, guaranteeing that individuals with ADHD are not only accommodated but also empowered to thrive in their careers. Through personalised gameplay, real-time performance monitoring and workplace integration, WADAQ stands out as a pioneering tool in bridging the gap between ADHD treatment and professional development.

9 Future works

Our future endeavours will include the integration of RL into WADAQ specifically for the purpose of enriching personalisation and adaptability. Future developments will involve refining user behaviour analysis, optimising gameplay adjustments and expanding the game library to address a broader range of ADHD symptoms and create more tailored experiences. To guarantee quality and real-world effectiveness, user testing will be conducted in workplaces with employees with ADHD. This approach will enable us to fine-tune the system's effectiveness in supporting individuals in work contexts as well as in improving focus, productivity and cognitive skills. Additionally, WADAQ will serve as a valuable employment tool, assisting organisations in understanding ADHD-related needs, optimising workplace accommodations and facilitating a structured, inclusive job interview process.

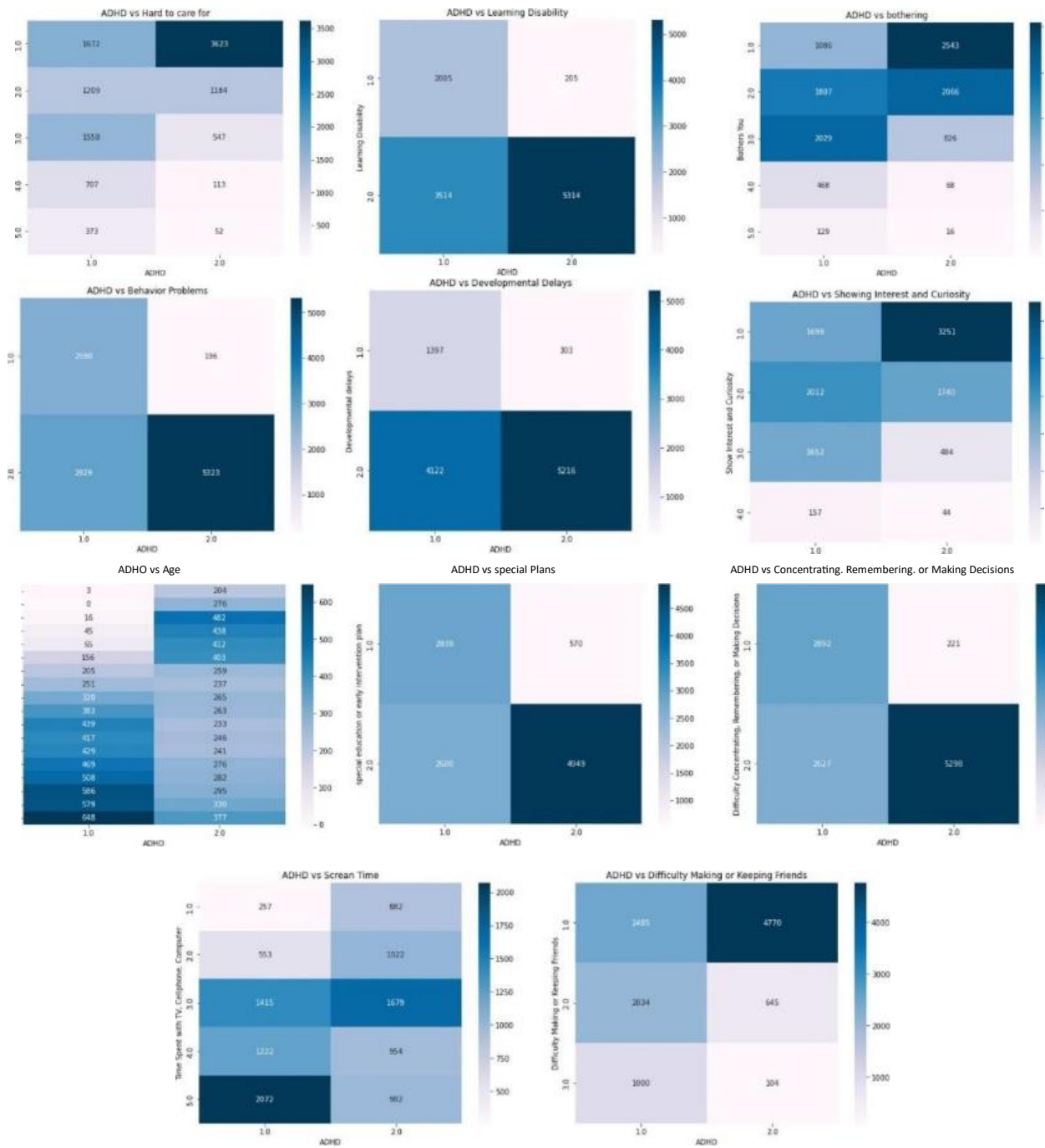
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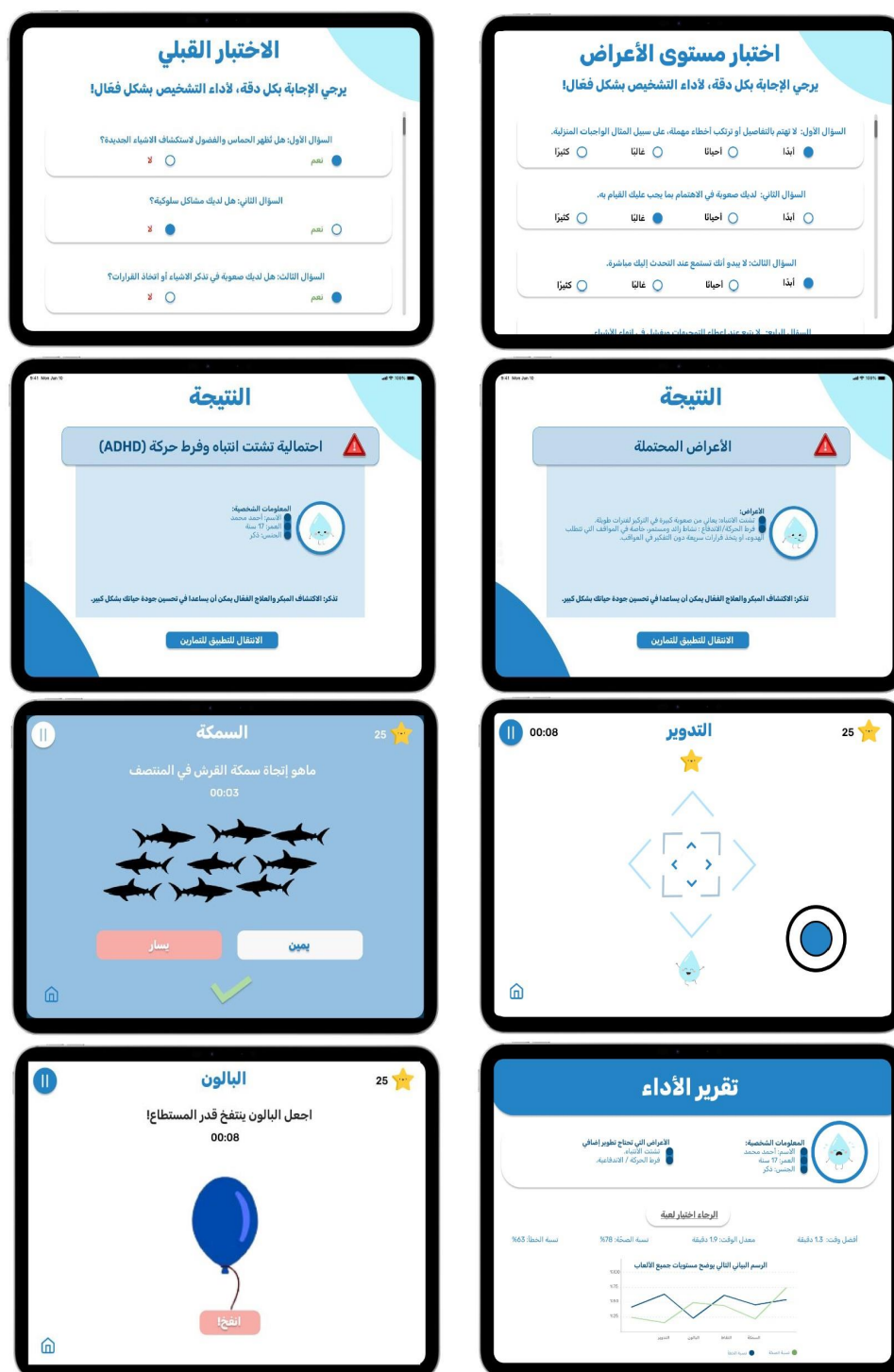
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APPENDICES



Appendix A: WADAQ post processing EDA



Appendix B: WADAQ application game Interfaces